Bahria University, Islamabad

Department of Software Engineering

DATABASE MANAGEMENT SYSTEM (SPRING-2025)

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LAB#05

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LAB

Proposal:

Bakery Order Placement Application

1. Introduction:

With the increasing demand for online ordering, bakeries need an efficient system to manage orders seamlessly. This project aims to develop a **Bakery Order Placement Application** that enables customers to place online orders for cakes and cookies, while also helping bakery staff manage inventory and order processing efficiently. The system will incorporate a **relational database** for effective data management.

2. Objectives:

The primary objectives of this project are:

- To provide an intuitive and user-friendly interface for customers to place orders.
- To automate order processing and inventory management for the bakery staff.
- To integrate a relational database for secure and efficient order tracking.

3. Features & Deliverables:

Customer Features:

- **User Authentication:** Secure login/signup functionality.
- **Order Placement:** Customers can select items, specify quantity, and provide delivery details.
- **Custom Orders:** An option to upload an image for personalized cake/cookie designs.
- **Shopping Cart:** Customers can add/remove items and view order summary before checkout.
- **Checkout & Payment:** A secure process for finalizing orders with payment integration.
- **Order Confirmation:** A confirmation message and order tracking details.

Admin & Bakery Staff Features:

- **Order Management:** View, update, and process customer orders.
- **Inventory Management:** Track stock levels, update product availability, and manage ingredients.
- **Customer Notes:** Special requests or instructions can be recorded.
- **Reporting & Analytics:** Insights into sales trends, order volume, and customer preferences.
- **Customer Support:** A contact section with company email and phone details.

4. Technology Stack:

- **Frontend:** HTML, CSS, JavaScript (Bootstrap for styling)
- **Backend:** Python (Flask)
- **Database:** MySQL (Flask-SQLAlchemy for ORM)
- **Tools:** VS Code, MySQL Workbench, Lucidchart, Visual Paradigm for database design

5. Database Design & Implementation:

Tables & Relationships:

- **Customers** (ID, Name, Email, Phone, Address, etc.)
- **Orders** (OrderID, CustomerID, OrderDate, TotalAmount, Status, etc.)
- **Products** (ProductID, Name, Price, StockQuantity, Category, etc.)
- **Payments** (PaymentID, OrderID, Amount, PaymentMethod, Status, etc.)

SQL Queries:

- Queries for **CRUD operations** (Create, Read, Update, Delete)
- Use of **joins** for relational data retrieval
- **Indexing & query optimization** for performance improvement

Database Normalization:

- Ensuring efficient structure with **minimal redundancy**

6. Project Timeline:

Phase	Task	Duration
Research & Planning	Requirements gathering, tech stack selection.	2 Weeks
Database Design	ERD creation, schema design.	2 Weeks
Development	UI design, backend logic, database integration.	4 Weeks
Testing & Debugging	Unit testing, performance optimization.	2 Weeks
Deployment & Report	Final deployment, documentation, and report writing.	2 Weeks

7. DBMS Report Breakdown:

The project report for the **Database Management System** will include:

- **Database Design:** Tables, relationships, constraints, and ER diagrams.
- **SQL Queries:** Queries for insertion, retrieval, updates, and deletions, along with optimization techniques.
- **Query Results:** Screenshots of executed queries and sample data output.
- **Data Integrity Measures:** Constraints and normalization techniques to ensure consistency.
- **UI Design:** Screenshots and descriptions of order placement and management.
- **Database Export File:** Providing an exportable file for easy database transition.

8. Expected Outcomes:

- A fully functional **Bakery Order Placement Application** with a seamless user experience.
- **Automated order and inventory management** to improve efficiency.
- A **robust database system** for secure and efficient data handling.
- A **well-documented project report** demonstrating OOP and DBMS concepts.

9. Conclusion:

This project will provide a **real-world solution** for bakeries by streamlining the order placement process while also improving inventory and customer management. Through the effective use of **database design principles**, this system will not only enhance customer experience but also optimize bakery operations.